

FEDERAL UNIVERSITY OF TECHNOLOGY MINNA, NIGER
STATE

SCHOOL OF ELECTRICAL ENGINEERING AND
TECHNOLOGY (SEET)

COURSE: EEE 415 (NUMERICAL METHODS II)

DEPARTMENT: ELECTRICAL/ELECTRONIC ENGINEERING

PRESENTATION ON NUMERICAL APPROXIMATION OF
THE ELECTROMAGNETIC FIELD INTEGRAL IN A
TRANSMISSION LINE BY GROUP 8

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EEE 415 – Numerical Methods II

Term Project: Problem 18

Numerical Approximation of the Electromagnetic Field Integral in a Transmission Line

1. Introduction and Problem Statement

Transmission lines carrying current produce surrounding magnetic fields that can lead to electromagnetic interference (EMI) in nearby circuits and communication systems.

To estimate magnetic field strength around a conductor, electromagnetic theory provides expressions that often involve definite integrals.

In practical engineering systems, the geometry of conductors and spacing between them can make analytical integration difficult. Numerical methods therefore become essential.

This project applies numerical integration techniques to evaluate a magnetic field integral using two numerical methods:

1. Composite Trapezoidal Rule
2. Composite Simpson's Rule

The accuracy and convergence of both methods are compared and the engineering implications for electromagnetic interference are discussed.

2. Mathematical Modelling

The magnetic field intensity around a long straight conductor carrying current I is given by:

$$H = I / (2\pi r)$$

Where:

H = magnetic field intensity (A/m)

I = current in the conductor (A)

r = distance from the conductor (m)

For general electromagnetic analysis the magnetic field may be expressed as:

$$H = (\mu_0 I / 4\pi) \int (1/r) dr$$

For this project the definite integral evaluated is:

$$\int (1/r) dr \text{ from } a \text{ to } b$$

Example limits used:

$a = 0.02 \text{ m}$

$b = 0.2 \text{ m}$

These limits represent distances measured outward from the conductor surface.

3. Numerical Method Selection

Two numerical integration methods are used in this study.

Trapezoidal Rule:

The integral $\int f(x)dx$ from a to b is approximated as

$$(h/2)[f(x_0) + 2f(x_1) + 2f(x_2) + \dots + f(x_n)]$$

where $h = (b-a)/n$

Characteristics:

- Simple to implement
- First-order accuracy
- Error proportional to h^2

Simpson's Rule:

Simpson's rule approximates the function using parabolic interpolation.

$$(h/3)[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \dots + f(x_n)]$$

Conditions:

- Number of intervals must be even

Accuracy:

- Higher accuracy than trapezoidal rule
- Error proportional to h^4

4. Algorithm Development

Algorithm for Trapezoidal Rule:

- Step 1: Define function $f(r) = 1/r$
- Step 2: Define limits a and b
- Step 3: Choose number of intervals n
- Step 4: Compute step size $h = (b-a)/n$
- Step 5: Compute first and last terms
- Step 6: Loop through interior points
- Step 7: Compute final integral value
- Step 8: Display result

Algorithm for Simpson's Rule:

Step 1: Define function
Step 2: Ensure n is even
Step 3: Compute step size
Step 4: Compute first and last terms
Step 5: Add odd indexed terms multiplied by 4
Step 6: Add even indexed terms multiplied by 2
Step 7: Compute final integral value

5. MATLAB Implementation

```
clc
clear
close all

% Function definition
f = @(r) 1./r;

% Limits
a = 0.02;
b = 0.2;

% Number of intervals
n = 10;

h = (b-a)/n;

x = a:h:b;
y = f(x);

%% Trapezoidal Rule
trap = y(1) + y(end);

for i = 2:n
    trap = trap + 2*y(i);
end

trap_result = (h/2)*trap;

%% Simpson Rule
if mod(n,2)~=0
    error('n must be even for Simpson rule')
end

simp = y(1) + y(end);

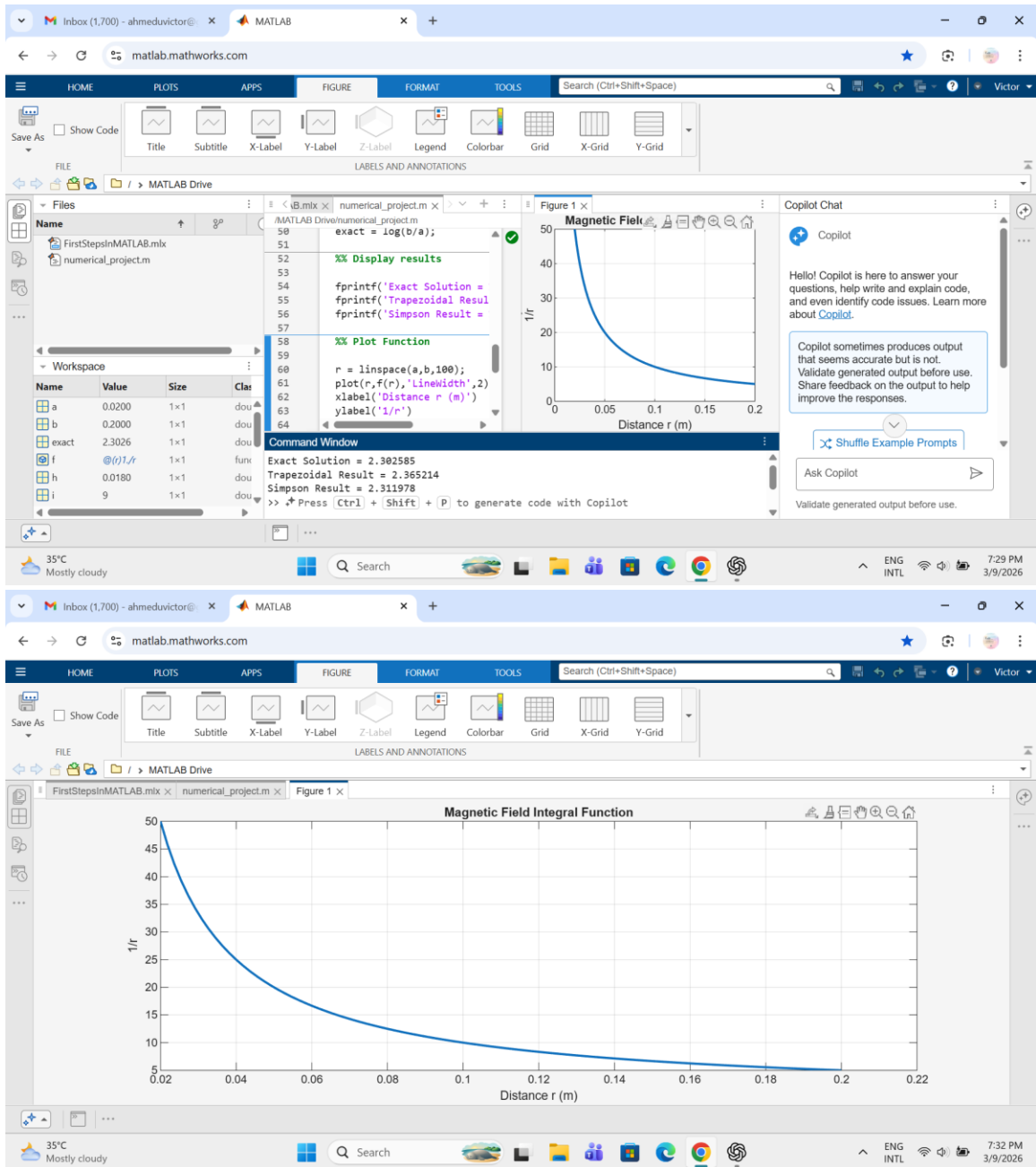
for i=2:2:n
    simp = simp + 4*y(i);
end
```

```
for i=3:2:n-1
    simp = simp + 2*y(i);
end

simp_result = (h/3)*simp;

exact = log(b/a);

fprintf('Exact Solution = %f\n',exact)
fprintf('Trapezoidal Result = %f\n',trap_result)
fprintf('Simpson Result = %f\n',simp_result)
```



NB; The attached images are the results obtained during the Matlab simulation

6. Error Analysis

The exact solution of the integral $\int (1/r) dr$ from a to b is:

$$\ln(b/a)$$

For:

$$a = 0.02$$

$$b = 0.2$$

Exact solution:

$$\ln(10) = 2.3026$$

The numerical results obtained using trapezoidal and Simpson's rules can be compared with this exact value to determine numerical error.

As the number of intervals increases, the error reduces. Simpson's rule typically converges faster than the trapezoidal rule.

7. Sensitivity Study

To study the effect of conductor spacing, the upper limit b can be varied:

$$b = 0.1 \text{ m}$$

$$b = 0.2 \text{ m}$$

$$b = 0.3 \text{ m}$$

The magnetic field is recalculated for each value and plotted against distance.

Observation:

As the distance from the conductor increases, the magnetic field intensity decreases.

8. Engineering Interpretation

Magnetic field strength decreases with distance according to the inverse relationship with r . Increasing spacing between transmission lines and nearby electronic equipment reduces electromagnetic interference.

Simpson's rule provides significantly higher numerical accuracy than the trapezoidal rule. For electromagnetic simulations requiring precision, Simpson's rule is preferable.

Accurate field estimation helps engineers design transmission corridors, position communication cables, and minimize EMI risks.

9. Conclusion

This project numerically evaluated the electromagnetic field integral around a transmission line conductor using trapezoidal and Simpson's rules.

The study showed that numerical integration methods are effective for electromagnetic field calculations. Simpson's rule demonstrated higher accuracy and faster convergence compared to the trapezoidal rule.

The analysis also confirmed that magnetic field strength decreases with increasing distance from the conductor.

10. Recommendations

Simpson's rule should be used when high accuracy is required for electromagnetic simulations. Trapezoidal rule may still be used for quick approximations.

Transmission line design should consider adequate spacing between conductors and nearby sensitive equipment to minimize electromagnetic interference.

11. References

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